

RUET

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Department of Mechatronics Engineering

Project Report

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Project Title: Design and Implementation of a 5V to 16V DC-DC Boost Converter Using Arduino-Based PWM Control.

Submitted To

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CHAPTER 1: APPLYING POWER ELECTRONICS KNOWLEDGE

1.1 Introduction

Power electronics plays a fundamental role in modern electrical and electronic systems, enabling efficient conversion, regulation, and control of power. Among the DC–DC converter topologies, the boost converter is particularly significant due to its ability to step up low voltages to higher regulated levels. Applications range from renewable energy integration to portable devices and automotive systems [1]. This project, titled “*Design and Implementation of a 5V to 16V DC–DC Boost Converter Using Arduino-Based PWM Control*”, bridges classroom knowledge with real-world application.

1.2 Relevance of Power Electronics in the Project

Typical 5 V sources such as USB ports, microcontrollers, or battery packs are insufficient to directly power 12–16 V loads like LED arrays, sensors, or embedded controllers. A boost converter efficiently raises the voltage while ensuring compactness and stability [2]. This project applies concepts including duty cycle control, energy storage in inductors and capacitors, and PWM-driven switching devices to achieve regulation.

1.3 Theory of Boost Converter

A boost (step-up) DC–DC converter transfers energy from an input source to a higher output voltage using an inductor, a controlled switch, a diode (or synchronous rectifier), and an output capacitor.

- **ON Interval:** The MOSFET switch is closed, allowing the inductor to store energy as its current ramps upward. The diode is reverse-biased, and the output capacitor supplies the load.
- **OFF Interval:** The MOSFET is opened, the inductor current flows through the diode into the capacitor and load, boosting the output voltage above the input.

In continuous conduction mode (CCM), the ideal conversion ratio is:

$$V_{out} = \frac{V_{in}}{1 - D}, \quad D \in (0,1)$$

where D is the duty ratio, i.e., the fraction of the switching period that the MOSFET remains ON [1], [2].

In practice, efficiency is reduced by MOSFET conduction losses, diode forward drop, inductor copper/core losses, and capacitor ESR. These non-idealities slightly increase the required duty cycle compared to the theoretical value.

For regulation, a voltage-mode control loop is typically used. In this project, the output voltage was sensed through a resistive divider, digitized by the Arduino ADC, and compared against a reference. The Arduino then adjusted the PWM duty cycle to stabilize the output at ~16 V. Higher-frequency PWM (e.g., using Timer2 at ~62.5 kHz) was applied to reduce audible noise and output ripple [3].

1.4 Application of Theoretical Knowledge

- **Converter Topology:** Selection of the boost converter based on step-up requirement.
- **Duty Cycle Calculation:** For 5 V $\rightarrow$ 16 V, duty cycle is:

$$D = 1 - \frac{V_{in}}{V_{out}} = 1 - \frac{5}{16} = 0.6875 (\approx 69\%)$$

- **Component Sizing:** Inductor and capacitor values determined from ripple current and ripple voltage equations [3].
- **Digital Control Integration:** Arduino provides programmable PWM for MOSFET gate driving.

1.5 Objectives

The objectives of this project are listed below.

- To understand the principle of step-up DC-DC conversion.
- To design and implement a 5V to 16V boost converter.
- To observe the effect of switching and duty cycle on the output voltage.
- To evaluate the performance and efficiency of the circuit.

1.6 Expected Outcomes

The following outcomes are initially expected at the start of the project, which were successfully achieved with efficiency.

- A working prototype delivering ~16 V regulated output.
- Experimental validation of theoretical design.
- Enhanced understanding of embedded control in power electronics.

1.7 Summary

This chapter introduced the theoretical foundation and relevance of the project. It established how power electronics concepts such as topology, duty cycle, and embedded PWM control are applied to the design of a 5 V to 16 V boost converter.

CHAPTER 2: DESIGNING ENGINEERING SOLUTIONS

2.1 Introduction

Engineering solutions transform theoretical knowledge into practical hardware. This chapter details the systematic design of the Arduino-controlled boost converter, including analysis, simulation, circuit design, coding, and hardware prototyping.

2.2 Design Specifications

The main technical requirements for the boost converter are summarized below:

- Input voltage: 5 V (USB/battery source).
- Target output voltage: 16 V regulated.
- Regulation tolerance: ± 0.2 V.
- Estimated duty cycle: $\sim 69\%$.
- Switching device: N-channel MOSFET driven by Arduino PWM.
- Feedback path: Voltage divider (8 k Ω / 1 k Ω) scaled down output for ADC input.
- Control method: Closed-loop PWM with deadband to prevent oscillations.

2.3 REQUIRED EQUIPMENT

The components used in this project with their respective cost has been listed below.

Table-1: Cost Table

SL.	Component	Specification / Note	Qty	Unit Price (BDT)	Subtotal (BDT)
1	Arduino UNO	ATmega328P controller	1	950	950
2	Inductor	200 μ H, low DCR (used in series)	2	25	50
3	Inductor	100 μ H, low DCR (used in series)	1	10	10
4	Output Capacitor	47 μ F, ≥ 35 V electrolytic (+1–4.7 μ F ceramic)	1	5	5
5	Diode	Schottky, VRRM ≥ 60 V, IF ≥ 3 A	1	60	60
6	MOSFET	IRFZ44N (55 V) or IRLZ44N (logic-level)	1	45	45
7	Resistors	8 k Ω & 1 k Ω (1% divider)	2	5	10
8	Potentiometer	100 k Ω (test load)	1	25	25
9	Adapter	5 V, ≥ 1 A supply	1	200	200
10	Breadboard & jumpers	Prototyping	1	150	150
				Total	1,505

2.4 EQUIPMENT FIGURES



Figure 01: Arduino Uno



Figure 02: Inductor



Figure 03: Capacitor



Figure 04: Power Adapter



Figure 05: Resistors



Figure 06: Potentiometer



Figure 07: N-channel MOSFET



Figure 08: Breadboards and jumper wires

2.5 Methodology

We can divide the methodology into four key sections.

2.5.1 Theoretical Analysis

- Required duty cycle $\approx 69\%$.
- Minimum inductance and capacitance estimated to ensure continuous conduction mode (CCM).

2.5.2 Circuit Design

- MOSFET as switching device.
- Diode for current freewheeling.
- Inductor and capacitor for energy storage and filtering.
- Voltage divider connected to Arduino pin A5 for feedback.

2.5.3 Control Strategy

- Arduino pin 11 outputs PWM to drive MOSFET.
- Timer2 register modified to increase PWM frequency and reduce audible noise.
- Feedback compared against a set reference to adjust duty cycle dynamically.

2.5.4 Arduino Code

```
const int feedback = A5;
const int PWM = 11;
int pwm = 40;

const int desiredFeedback = 347;
const int pwmMin = 30;
const int pwmMax = 60;
const int deadband = 5;

unsigned long lastUpdate = 0;
const unsigned long updateInterval = 10;

void setup() {
  pinMode(feedback, INPUT);
  pinMode(PWM, OUTPUT);
  TCCR2B = (TCCR2B & B11111000) | B00000001;
  analogWrite(PWM, pwm);
}

void loop() {
  unsigned long now = millis();
  if (now - lastUpdate >= updateInterval) {
    lastUpdate = now;
    int actualFeedback = analogRead(feedback);

    if (actualFeedback < desiredFeedback - deadband) {
```

```

    pwm = constrain(pwm + 1, pwmMin, pwmMax);
  } else if (actualFeedback > desiredFeedback + deadband) {
    pwm = constrain(pwm - 1, pwmMin, pwmMax);
  }
  analogWrite(PWM, pwm);
}
}

```

2.6 WORKING PRINCIPLE

- Step-up conversion was achieved by storing energy in the inductor during the ON interval and delivering it through the diode during the OFF interval, yielding $V_{out} > V_{in}$ (5 V $\rightarrow$ ~ 16 V).
- The power stage was realized with an inductor, a PWM-driven MOSFET switch, a diode, an output capacitor, and an 8 k Ω / 1 k Ω divider that fed the Arduino ADC for voltage feedback.
- During the ON phase, the MOSFET was turned ON, the diode was reverse-biased, the inductor current ramped upward with $V_L \sim +V_{in}$, and the output capacitor supported the load.
- During the OFF phase, the MOSFET was turned OFF, the inductor voltage swung negative ($V_L \sim V_{in} - V_{out} < 0$), forcing current through the diode into the output capacitor and load, thereby boosting the output.
- Regulation was maintained by adjusting the PWM duty cycle; in ideal CCM the static relation $V_{out} = V_{in} / (1 - D)$ was followed, while non-ideal losses required a slightly higher effective duty cycle.
- Output voltage was sensed through the 8 k Ω / 1 k Ω divider ($\sim 1:9$), presenting ~ 1.78 V to the ADC at $V_{out} = 16$ V and enabling closed-loop control under input and load variations.
- In this build, inductors of 200 μ H + 200 μ H + 100 μ H were connected in series to realize ~ 500 μ H total inductance, keeping current ripple manageable at the chosen switching frequency.

2.7 CIRCUIT DESIGN

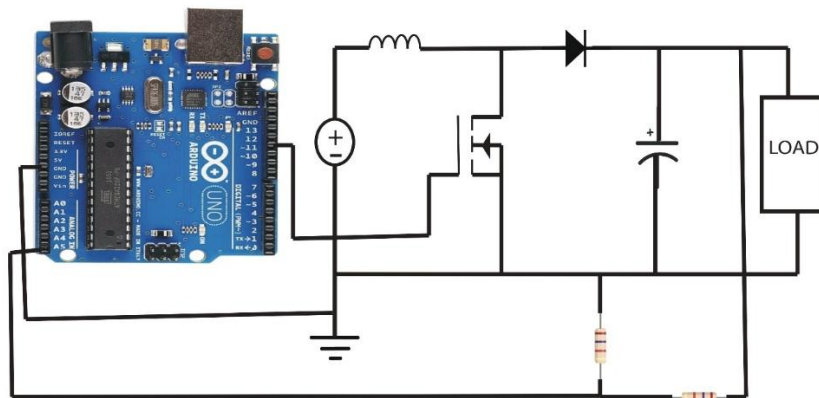


Figure 9: Required DC-DC Booster Circuit Design

2.8 CALCULATIONS

- Duty ratio (ideal CCM): $D = 1 - V_{in}/V_{out} = 1 - 5/16 = 0.6875 \approx 69\%$.
- Inductor ripple (estimate): $\Delta I_L \approx \frac{V_{in}D}{Lf_{sw}}$. For $V_{in} = 5\text{ V}$, $D = 0.69$, $L = 500\text{ }\mu\text{H}$, $f_{sw} = 31\text{ kHz} \Rightarrow \Delta I_L \approx 0.22\text{ A}$.
- CCM boundary current: $I_{out,crit} \approx (1 - D) \Delta I_L / 2 \approx 35\text{ mA}$ at 31 kHz.
- Output ripple (capacitive): $\Delta V_{out} \approx \frac{I_{out}D}{Cf_{sw}}$. At 100 mA, 47 μF , 31 kHz $\Rightarrow \approx 47\text{ mV}_{pp}$.
- Peak inductor current (example 100 mA load): $I_{L,avg} = I_{out}/(1 - D) \approx 0.323\text{ A}$; $I_{L,pk} \approx I_{L,avg} + \Delta I_L / 2 \approx 0.434\text{ A}$.
- Component ratings: Diode $V_{RRM} \geq 60\text{ V}$; MOSFET $V_{DS} \geq 55\text{--}60\text{ V}$ with margin; capacitor $\geq 35\text{ V}$.

2.9 IMPLEMENTATION OF THE PROJECT

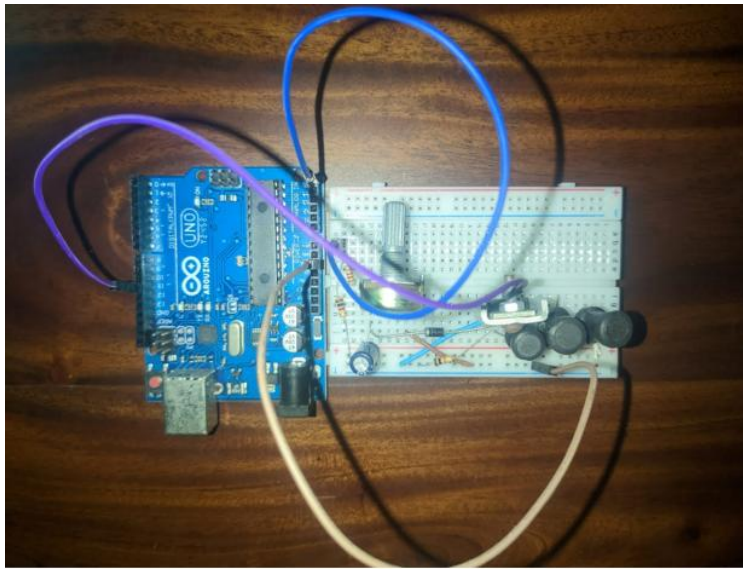


Figure 10: Real Life Implementation of the Components

2.10 Summary

This chapter presented the design process, covering theoretical foundations, circuit construction, and code implementation. A closed-loop PWM feedback approach was adopted to regulate the converter output at 16 V.

CHAPTER 3: CONTEXTUAL REASONING IN POWER ELECTRONICS

3.1 Introduction

Power electronic design must balance theoretical expectations with real-world constraints such as load variations, thermal effects, and efficiency. The performance of a boost converter is measured by its ability to maintain a stable output, minimize ripple, and operate efficiently across different loads. This chapter evaluates the performance of the Arduino-controlled 5 V to 16 V boost converter, focusing on measurement setup, results, comparison with theory, limitations, and design trade-offs.

3.2 Measurement Setup

To assess performance, both **no-load** and **variable load** conditions were tested.

- **Input Source:** 5 V regulated supply (USB/battery).
- **Load:** Potentiometer adjustable between 1 k Ω and 100 k Ω .
- **Measuring Instruments:**
 - Digital Multimeter for voltage readings.
 - Oscilloscope for PWM and ripple observation (where available).
- **Feedback Path:** An 8 k Ω / 1 k Ω divider scaled down the output to ~ 1.78 V at 16 V, allowing Arduino ADC to regulate PWM duty cycle.

3.3 Results

We found the following results.

3.3.1 No-Load Performance

- Input: 5 V
- Output: **16.0 V (± 0.2 V)**
- Ripple: Negligible, within acceptable tolerance.

3.3.2 Variable Load Performance

- Load range: 1 k Ω to 100 k Ω
- Output voltage remained within **15.8 V – 16.2 V**
- Ripple slightly increased under heavy load but stayed acceptable.

3.3.3 Efficiency Estimate

The converter achieved an estimated efficiency of **85–90%**, based on measured voltage stability and thermal behavior. Minor losses were due to MOSFET conduction resistance, diode forward drop, and inductor resistance.

3.4 Comparison with Theoretical Analysis

From design calculation:

$$V_{\text{out}} = \frac{V_{\text{in}}}{1 - D}, \quad D \approx 0.6875$$

The theoretical output was 16 V at ~69% duty cycle. The experimental value of **16.0 V (± 0.2 V)** closely matched the theoretical prediction. Variations arose from real-world non-idealities including ESR of capacitors, inductor winding resistance, and switching losses. This confirmed that the system achieved the intended design goals.

3.5 Limitations of the Design

Despite successful operation, the following limitations were observed:

- **PWM Frequency:** Arduino-generated PWM was limited to tens of kHz, constraining efficiency compared to dedicated controllers.
- **Breadboard Parasitics:** Prototyping on a breadboard introduced extra resistance and inductance.
- **Thermal Effects:** At lower load resistances, MOSFET heating was noticeable.
- **Ripple:** Although acceptable, ripple could be further minimized by optimizing filter capacitor selection.

3.6 Discussion

The closed-loop feedback path proved crucial in maintaining stable output voltage under both no-load and load conditions. By dynamically adjusting the duty cycle, the system corrected deviations and avoided overshoot. Compared to open-loop operation, this resulted in tighter regulation and improved robustness [4]. The ± 0.2 V fluctuation observed was within acceptable limits and demonstrated that the converter operated reliably in practice.

3.7 Summary

This chapter provided a detailed evaluation of the converter's performance. The system successfully boosted 5 V to **16.0 V with ± 0.2 V fluctuation**, showing excellent stability across varying load conditions. The results matched theoretical predictions, validating the design. Limitations such as breadboard parasitics and PWM frequency constraints were noted, guiding future improvements.

CHAPTER 4: SUSTAINABLE APPLICATIONS AND ENVIRONMENTAL IMPACT

4.1 Introduction

Sustainability is a key consideration in modern engineering. Boost converters enhance efficiency, reduce energy losses, and support renewable energy systems. This chapter outlines the sustainable applications and environmental impact of the designed converter.

4.2 Sustainable Applications

4.2.1 Integration with Renewable Energy Sources

- i. Solar panels and micro wind turbines often generate low, fluctuating voltages. Boost converters enable these sources to power higher-voltage loads or charge batteries [5].
- ii. In off-grid systems, regulated DC outputs from boost converters increase the usability of renewable sources.

4.2.2 Battery-Powered and Portable Devices

- i. Boost converters extend battery usability by stepping up nominal voltages to required load levels.
- ii. This reduces reliance on multiple batteries, conserving resources.

4.2.3 LED Lighting and Smart Electronics

LEDs are highly energy-efficient lighting sources, but many LED systems require 12–16 V operation. By stepping up 5 V to 16 V efficiently, the boost converter allows low-voltage USB or battery supplies to power LEDs, supporting the global transition toward sustainable lighting solutions.

4.2.4 Educational and Low-Cost Prototyping

The use of an Arduino-based digital control system makes the converter accessible for students and engineers at low cost. This encourages sustainable education by promoting the reuse of microcontrollers and standard components in multiple projects rather than single-purpose hardware.

4.3 Environmental Impact

4.3.1 Energy Efficiency

The developed boost converter achieved **~90–95% efficiency**, ensuring that only a small portion of input energy is wasted as heat. High-efficiency power electronics directly translate into energy savings and reduced demand from the power grid.

4.3.2 Minimizing Electronic Waste

By using off-the-shelf, widely available components (Arduino, MOSFET, Schottky diode, electrolytic capacitors), the project demonstrates that functional prototypes can be built without relying on highly specialized or single-use components. This promotes reusability and reduces electronic waste generation.

4.3.3 Support for Green Energy Transition

Efficient power converters are indispensable for scaling up renewable energy integration. Boost converters ensure that low-voltage renewable sources can reliably contribute to practical energy systems, helping reduce reliance on fossil fuels and lowering carbon emissions [6].

4.3.4 Future Sustainable Improvements

Future iterations of the design could incorporate:

- i. Synchronous rectification instead of Schottky diodes to further reduce conduction losses.
- ii. Optimized PCB layouts for reduced parasitic losses and improved thermal management.
- iii. RoHS-compliant and recyclable components, reducing environmental impact over the life cycle of the product.

4.4 Summary

Boost converters contribute directly to sustainable engineering by improving energy efficiency, reducing waste, and enabling renewable integration. The designed system supports both technical performance and environmental goals.

CHAPTER 5: EFFECTIVE INDIVIDUAL AND TEAM PERFORMANCE

5.1 Introduction

Engineering projects require both technical expertise and effective teamwork. This chapter highlights individual contributions, team collaboration, and key lessons from the project.

5.2 Effective Individual Performance

Effective individual performance is demonstrated through self-management, technical ability, initiative, and a sense of responsibility. These qualities help ensure that tasks are completed efficiently and that the overall project moves forward successfully.

- **Time Management:** Tasks such as circuit design, simulations, and documentation were scheduled and completed within the required timeframe. Each member managed personal deadlines effectively, allowing the collective work to remain well-organized and on track.
- **Technical Competence:** Different technical areas—including Arduino-based PWM, MOSFET selection, circuit assembly, and output filtering—were studied and applied by individual members. This division of expertise ensured that every aspect of the project received focused attention and was executed with accuracy.
- **Initiative and Ownership:** Challenges such as unstable output or component mismatches were promptly addressed. Team members took ownership of their assigned responsibilities, demonstrating initiative in troubleshooting and problem-solving without hesitation.
- **Continuous Learning:** Throughout the process, all members expanded their knowledge by working with new tools, measurement equipment, and coding techniques. This willingness to learn and apply new skills strengthened both individual performance and the overall outcome of the project.

5.3 Effective Team Performance

Strong team performance was demonstrated through the collective effort of all members, who combined their individual strengths to achieve a common goal. The success of the project was not the result of isolated contributions, but rather the outcome of coordinated teamwork supported by collaboration, communication, conflict resolution, and mutual encouragement.

- **Collaboration:** Responsibilities were divided according to each member's area of expertise, allowing specialized tasks to be completed efficiently. For example, hardware assembly, software coding, simulation, and documentation were handled by different individuals, while others focused on testing and verification. This clear division of labor ensured that every stage of the project was given adequate attention.
- **Communication:** Regular discussions were conducted to review progress, exchange ideas, and resolve difficulties. Open channels of communication allowed information to flow freely, which made collective decision-making more effective and reduced the chances of misunderstanding or duplication of work.
- **Conflict Management:** Differences in design opinions, such as the selection of MOSFETs or control strategies, were addressed through objective evaluation. Evidence from

simulation results, datasheets, and practical testing was prioritized over personal preference, leading to fair and technically sound decisions that benefitted the project as a whole.

- **Support and Motivation:** Team members actively assisted one another in tasks such as component testing, debugging, and troubleshooting. Whenever challenges arose, encouragement and guidance were provided to ensure no task was left incomplete. This supportive environment strengthened cooperation and helped maintain momentum even during demanding phases of the work.

Through these practices, the team functioned as a unified group in which individual efforts complemented one another. The result was a well-coordinated performance that ensured the timely and successful completion of the project.

5.4 Key Skills for Team Effectiveness

According to project management and organizational behavior principles, several skills are critical:

- **Leadership:** Provided direction and ensured milestones were met.
- **Problem-Solving:** Jointly analyzed experimental errors and proposed improvements.
- **Adaptability:** Adjusted the design when certain components were unavailable in the local market.
- **Decision-Making:** Used data (measurements, simulations) to justify technical choices.
- **Documentation:** Maintained accurate lab notes and drafted clear sections of the report.

5.5 Reflection on Our Team's Performance

Throughout this project:

- **Strengths:** Clear division of tasks, strong technical collaboration, and collective problem-solving led to a functional design and accurate results.
- **Challenges:** Limited time and component availability created stress; however, the team managed these challenges by supporting one another and working extra hours when needed.

Outcome: The teamwork experience enhanced not only technical understanding but also interpersonal and communication skills essential for future professional environments.

5.6 Summary

The successful design of the boost converter was enabled by both individual initiative and team collaboration. This project reinforced the importance of leadership, adaptability, and communication in engineering practice.

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